(a) [Series solution]

Assume a separated solution : $u(x,t) = X(x)\,T(t)$

Plugging it into the wave equation, we get :

$$X(x)\,T''(t) = c^2 X''(x)\,T(t)$$

Dividing by $-c^2XT$, we have : $-\frac{T''}{c^2T} = -\frac{X''}{X} = \lambda$.

Since $\frac{\partial\lambda}{\partial x} = \frac{\partial\lambda}{\partial t} = 0$, λ must be a constant.

Case 1. If $\lambda = 0$, then $X''(x) = 0$, which means that $X = C + Dx$.
Notice that $X(0) = X(l) = 0$, so $C = D = 0$.
Then $X \equiv 0$. So does $u(x,t)$. It is a contradiction.

Case 2. If $\lambda < 0$, then $\lambda = -\gamma^2$, which means that $X(x) = Ce^{\gamma x} + De^{-\gamma x}$.
Notice that $X(0) = X(l) = 0$, so $C = D = 0$
Then $X \equiv 0$. So does $u(x,t)$. It is a contradiction.

Case 3. If λ is a complex number. Let the complex number γ be the root of $-\lambda$. Then $X(x) = Ce^{\gamma x} + De^{-\gamma x}$
Notice that $X(0) = X(l) = 0$. So $C + D = 0 = Ce^{\gamma l} + De^{-\gamma l}$.
Therefore $e^{2\gamma l} = 1$, which means $\mathrm{Re}(\gamma) = 0$ and $2l \cdot \mathrm{Im}(\gamma) = 2\pi n$ where n is an integer. Thus $\lambda = -\gamma^2 = \frac{n^2\pi^2}{l^2}$ is real and positive. It is a contradiction.

Thus, the only case is $\lambda > 0$

So, let $\lambda=\beta^2$ where $\beta>0$. Then $X(x)$ and $T(t)$ satisfy:

$$X''(x)+\beta^2X(x)=0 \quad \text{and} \quad T''(t)+c^2\beta^2T(t)=0$$

The solutions have the form:

$$\begin{cases} X(x)=C\cos\beta x+D\sin\beta x \\ T(t)=A\cos\beta ct+B\sin\beta ct \end{cases}$$

where A, B, C, D are constants.

Now consider the boundary conditions $0=X(0)=C$ and $0=X(L)=D\sin\beta L$

Thus $\beta L=n\pi$. Then summing all the possible n together:

$$u(x,t)=\sum_n\left(A_n\cos\frac{n\pi ct}{L}+B_n\sin\frac{n\pi ct}{L}\right)\sin\frac{n\pi x}{L}$$

For the initial data we can get:

$$\begin{cases} \phi(x)=u(x,0)=\sum_n A_n\sin\frac{n\pi x}{L} \\ \psi(x)=u_t(x,0)=\sum_n\frac{n\pi c}{L}B_n\sin\frac{n\pi x}{L} \end{cases}$$

which are Fourier sine series.

From hint we know that $\int_0^L\sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}dx=\frac{L}{2}\delta_{n,m}$

So. $\int_0^L\phi(x)\sin\frac{m\pi x}{L}dx=\int_0^L\sum_n A_n\sin\frac{n\pi x}{L}\cdot\sin\frac{m\pi x}{L}dx=\sum_n A_n\int_0^L\sin\frac{n\pi x}{L}\cdot\sin\frac{m\pi x}{L}dx$

$=A_n\cdot\frac{L}{2}$ when $n=m$.

Thus $A_n=\frac{2}{L}\int_0^L\phi(x)\sin\frac{m\pi x}{L}dx$.

Similarly, we can get $B_n=\frac{2}{n\pi c}\int_0^L\psi(x)\sin\frac{n\pi x}{L}dx$.

(b) [Series solution]

Assume a separated solution: $u(x,t) = X(x)T(t)$

Plugging it into the heat equation, we get:

$$\frac{T'}{a^2T} = \frac{X''}{X} = -\lambda = \text{constant}$$

From $\frac{T'}{a^2T} = -\lambda$ we can get: $T(t) = Ae^{-\lambda a^2 t}$.

Just like (a), we should discuss λ:

① $\lambda = 0$ ...

② $\lambda < 0$ ...

③ λ is a complex number ...

Thus, the only case is $\lambda > 0$

Let $\lambda = \beta^2$ where $\beta > 0$, we get $X''(x) + \beta^2 X(x) = 0$

The solutions have the form: $X(x) = C\cos\beta x + D\sin\beta x$

Consider the boundary conditions $X(0) = 0 = C$, $X(L) = 0 = D\cdot\sin\beta x = D\cdot\sin\beta L$

Thus $\beta L = n\pi$. Then summing all the possible n together, we have:

$$u(x,t) = \sum_{n=1}^{\infty} A_n e^{-\frac{n^2\pi^2a^2}{L^2}t} \sin\frac{n\pi x}{L}$$

For the initial data, $u(x,0) = \phi(x) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{L}$

From what we know that $\int_0^L \sin\frac{n\pi x}{L}\cdot\sin\frac{m\pi x}{L}\,dx = \frac{L}{2}\delta_{n,m}$

So, $\int_0^L \phi(x)\sin\frac{m\pi x}{L}\,dx = \int_0^L \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{L}\cdot\sin\frac{m\pi x}{L}\,dx = \sum_{n=1}^{\infty} A_n \int_0^L \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\,dx$

$= A_n\cdot\frac{L}{2}$ when $n = m$

Thus, $A_n = \frac{2}{L}\int_0^L \phi(x)\sin\frac{n\pi x}{L}\,dx$